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Final Project Specification

First, congratulations are in order for completing modules 2-4 of this course! If you followed along and completed all the exercises, you are now experienced with fundamental data, signal, and image analysis tools in MATLAB. You may have also noticed that along the way that you have developed a number of useful MATLAB functions that you can keep and use later. I never delete a function, just like I never throw away a tool in my garage. I may not anticipate needing the tool in the future, but eventually it seems inevitable that every tool becomes critically useful again at some point for some project. This final project requires using a few of the existing and custom functions you have learned about along the way.

Object localization is the process of recognizing and locating one or more type of object in an image. In this final project, you are being tasked to develop an image analysis-based solution for counting coin change for a bank. A camera is placed above the bank teller's countertop which will photograph coins placed on the countertop. When someone spreads any combination of quarters, nickels, and dimes on the countertop, your program will analyze an image taken from the camera to: (1) localize the coins, (2) classify each coin as a quarter, nickel, or dime, (3) add up the total value of the coins in the image.

An outline of the steps needed in the script is as follows:

1. Localize the centroid of each coin
2. Measure features for each coin using a so-called "filter bank" of matching filters
3. Perform k-means clustering of features for an unsupervised learning coin-type classifier, and visualize the results

The following 5 assessment items break this project up into a sequence of steps. Complete each item to complete your final project and the course. Start with the Final project script 1 and go through the sequence of outlined steps in order. Final project script 2 will become available upon completing script 1, and script 3 will become available upon completing script 2. Two of the scripts require writing functions. Once you reach the steps where these functions are needed, save and exit to the corresponding assessment that asks you to create the needed function. When finished with that assessment, copy and paste your function into the bottom of the corresponding final project script to embed it in the script, permitting it to be used in your script.

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